

Solids

Crystalline solids with N atoms can be shown to be a collection of $3N$ independent one-dimensional harmonic oscillators.

These $3N$ harmonic oscillators are known as the $3N$ normal modes of the crystalline solid.

The Hamiltonian for the solid can be written as,

$$H = \sum_i \frac{1}{2} (\dot{q}_i^2 + \omega_i^2 q_i^2) .$$

When these oscillators are quantized, the total energy can be written as,

$$E = \sum_i (n_i + 1/2) \hbar \omega_i$$

These quanta of vibrations are known as **phonons**.

When the normal modes are treated classically, from the equipartition theorem,

$$U = 6N \left(\frac{1}{2} k_B T \right) = 3N k_B T$$

$$\Rightarrow \text{Molar specific heat capacity, } C_v = \left. \frac{\partial U}{\partial T} \right|_{N=N_A} = 3R$$

This result is known as the Dulong-Petit Law.

Einstein's Treatment

Assume that all the vibrational modes have the same energy ω_E .

∴ The partition function of a single oscillator is given by

$$\begin{aligned} Z^{(1)} &= e^{-\beta \hbar \omega_E / 2} + e^{-3\beta \hbar \omega_E / 2} + e^{-5\beta \hbar \omega_E / 2} + \dots \\ &= \frac{e^{-\beta \hbar \omega_E / 2}}{1 - e^{-\beta \hbar \omega_E}} = \frac{e^{\beta \hbar \omega_E / 2}}{e^{\beta \hbar \omega_E} - 1} \end{aligned}$$

$$\Rightarrow U^{(1)} = - \frac{\partial \ln Z^{(1)}}{\partial \beta} = \frac{\partial}{\partial \beta} \left[\ln (e^{\beta \hbar \omega_E} - 1) - \frac{\beta \hbar \omega_E}{2} \right]$$

$$U^{(1)} = \frac{\hbar \omega_E e^{\beta \hbar \omega_E}}{e^{\beta \hbar \omega_E} - 1} - \frac{\hbar \omega_E}{2} = \frac{\hbar \omega_E}{e^{\beta \hbar \omega_E} - 1} + \frac{\hbar \omega_E}{2}$$

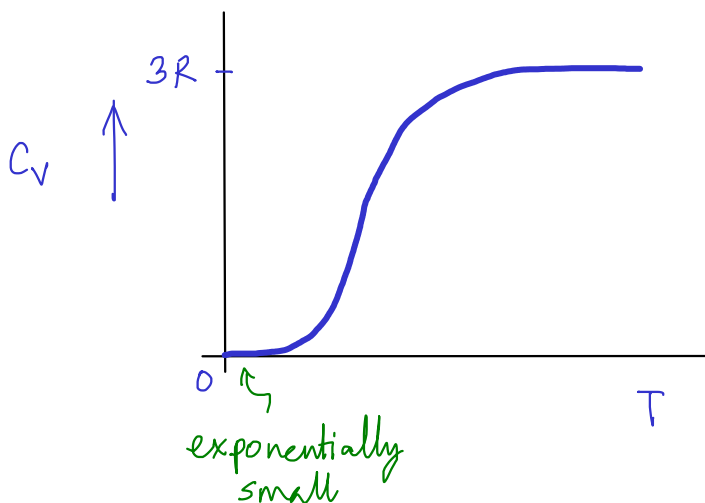
$$\Rightarrow U = 3N U^{(1)} = 3N \left[\frac{\hbar \omega_E}{e^{\beta \hbar \omega_E} - 1} + \frac{\hbar \omega_E}{2} \right]$$

$$\Rightarrow C_V = 3N_A \hbar \omega_E \frac{e^{\beta \hbar \omega_E} \hbar \omega_E / k_B T^2}{(e^{\beta \hbar \omega_E} - 1)^2}$$

$$= \frac{(\beta \hbar \omega_E)^2 e^{\beta \hbar \omega_E}}{(e^{\beta \hbar \omega_E} - 1)^2} 3R$$

Define the Einstein temperature, Θ_E using $k_B \Theta_E = \hbar \omega_E$.

$$\Rightarrow C_V = \frac{(\Theta_E/T)^2 e^{\Theta_E/T}}{(e^{\Theta_E/T} - 1)^2} 3R \quad \begin{cases} \approx 3R & \text{if } T \gg \Theta_E \text{ (Dulong-Petit)} \\ \approx (\Theta_E/T)^2 e^{-\Theta_E/T} 3R & \text{if } T \ll \Theta_E \end{cases}$$



Debye's treatment

Assume $\omega(\vec{k}) = c_s k$ where c_s is the speed of sound.

There are three such modes for each value of $\vec{k}$ — acoustic modes.

Also, consider a spherical region with $|\vec{k}| < k_D$ to participate in the vibrations of the solid.

The total volume in k -space is given by $\frac{4}{3}\pi k_D^3$

The volume associated with a single value of $\vec{k}$ is $\left(\frac{2\pi}{L}\right)^3 = \frac{(2\pi)^3}{V}$

∴ For a lattice with N points, there are N points in the first Brillouin zone in k -space.

⇒ Total volume of points in the first Brillouin zone = $N \frac{(2\pi)^3}{V} = (2\pi)^3 n$

Matching the number of points in the Debye sphere with the number of points in the first Brillouin zone, we have,

$$\Rightarrow n = \frac{k_D^3}{6\pi^2} \quad \text{and} \quad V = \frac{6\pi^2 N}{k_D^3}$$

$$\therefore U = 3 \sum_{\vec{k} \in \text{BZ}} \hbar \omega(\vec{k}) \left(n_{\vec{k}} + \frac{1}{2} \right)$$

$$= \frac{3V}{(2\pi)^3} \int_0^{k_D} 4\pi k^2 dk \hbar c_s k \left(\frac{1}{e^{\beta \hbar c_s k} - 1} + \frac{1}{2} \right) \quad \text{--- (1)}$$

$$= \frac{3V}{2\pi^2} \frac{\hbar c_s}{(\beta \hbar c_s)^4} \int_0^{\beta \hbar c_s k_D} \frac{x^3 dx}{e^x - 1} + \frac{3V}{4\pi^2} \hbar c_s \frac{k_D^4}{4} \quad x = \beta \hbar c_s k$$

Let $k_B \Theta_D = \hbar c_s k_D = \hbar \omega_D \quad \leftarrow \text{Definitions of Debye temperature } (\Theta_D) \text{ and Debye frequency } \omega_D.$

$$\Rightarrow \beta \hbar c_s k_D = \Theta_D / T$$

$$U = 3 \frac{6\pi^2 N}{2\pi^2 k_D^3} \hbar c_s k_D^4 \left(\frac{T}{\Theta_D} \right)^4 \int_0^{\Theta_D/T} \frac{x^3 dx}{e^x - 1} + \frac{3V}{16\pi^2} k_B \Theta_D 6\pi^2 n$$

$$= 9N k_B \frac{T^4}{(\Theta_D)^3} \int_0^{\Theta_D/T} \frac{x^3 dx}{e^x - 1} + \frac{9N k_B \Theta_D}{8} \quad \text{--- (2)}$$

At low temperatures, in the first term, the upper limit of the integral can be replaced by infinity.

Using, $\int_0^{\infty} \frac{x^3 dx}{e^x - 1} = \frac{\pi^4}{15}$, we get,

$$U \simeq \frac{3Nk_B}{5} \frac{\pi^4}{\Theta_D^3} T^4 + \frac{9Nk_B \Theta_D}{8}$$

$\Rightarrow$ Molar specific heat capacity at low temperatures is

$$C_V \simeq \frac{12\pi^4}{5} \left(\frac{T}{\Theta_D} \right)^3 R$$

This is much bigger than the Einstein's result (which is exponentially small at low temperatures, and gives the correct temperature for modes for which $\omega(k) \propto k$ as $k \rightarrow 0$, i.e. the acoustic modes.

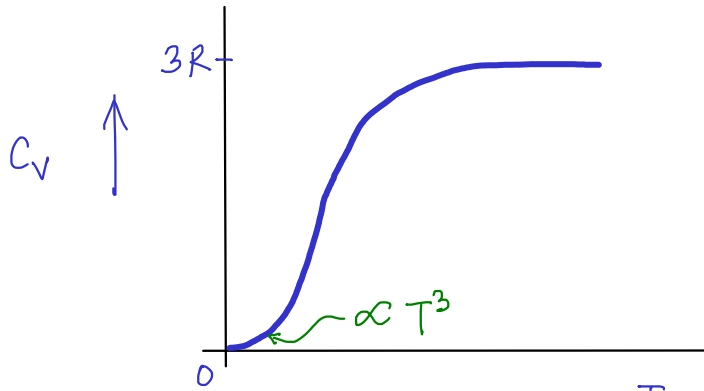
For modes for which $\omega(k) \rightarrow \text{constant} > 0$ for $k \rightarrow 0$, Einstein's description works.

High temperature result from Debye theory:

From Eq. (1), as $T \rightarrow \infty$, $\beta \rightarrow 0$, $e^{\beta \hbar c_s k} - 1 \rightarrow \beta \hbar c_s k$

$$\begin{aligned} \Rightarrow U &\simeq \frac{3V}{(2\pi)^3} \int_0^{k_D} 4\pi k^2 dk \hbar c_s k \left(\frac{1}{e^{\beta \hbar c_s k} - 1} \right) + \text{constant} \\ &\simeq \frac{3V}{(2\pi)^3} \int_0^{k_D} 4\pi k^2 dk k_B T + \text{constant} \\ &= \frac{3V}{(2\pi)^3} \frac{4}{3} \pi k_D^3 k_B T + \text{constant} = 3Nk_B T + \text{constant} \end{aligned}$$

$\Rightarrow C_V \simeq 3R$ as $T \rightarrow \infty$ (Dulong-Petit result)



General expression for c_v using Debye's approach

From Eq (2), we get

$$C_V = 36 R \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} \frac{x^3 dx}{e^x - 1} + 9R \frac{T^4}{\Theta_D^3} \frac{(\Theta_D/T)^3}{e^{\Theta_D/T} - 1} \left(-\frac{\Theta_D}{T^2} \right)$$

$$= 36 R \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} \frac{x^3 dx}{e^x - 1} - 9R \frac{\Theta_D/T}{e^{\Theta_D/T} - 1}$$

